Lecture 11

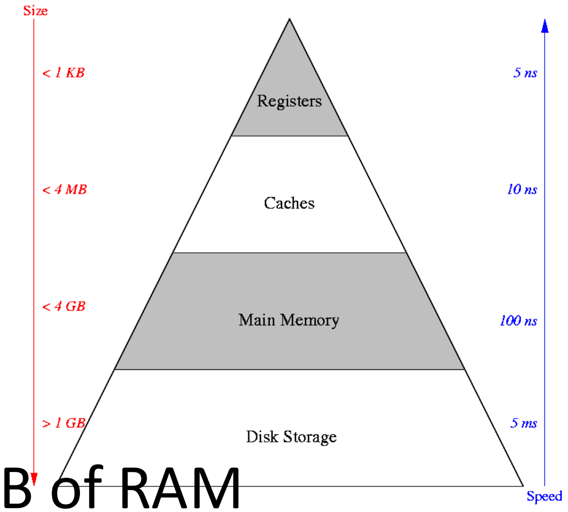
Virtual Memory

Some pictures from work of David Black Schaffer

Previously ..

- Locality of Reference
  - Spatial and Temporal
- Cache hit and miss
- Mapping Function
  - Direct, Associative, and Set-associative mapping
- Replacement Algorithm
  - LRU

Initiate with Some Problems



(1) Not enough RAM

- 32 bit address space, if we don’t have 4GB of RAM

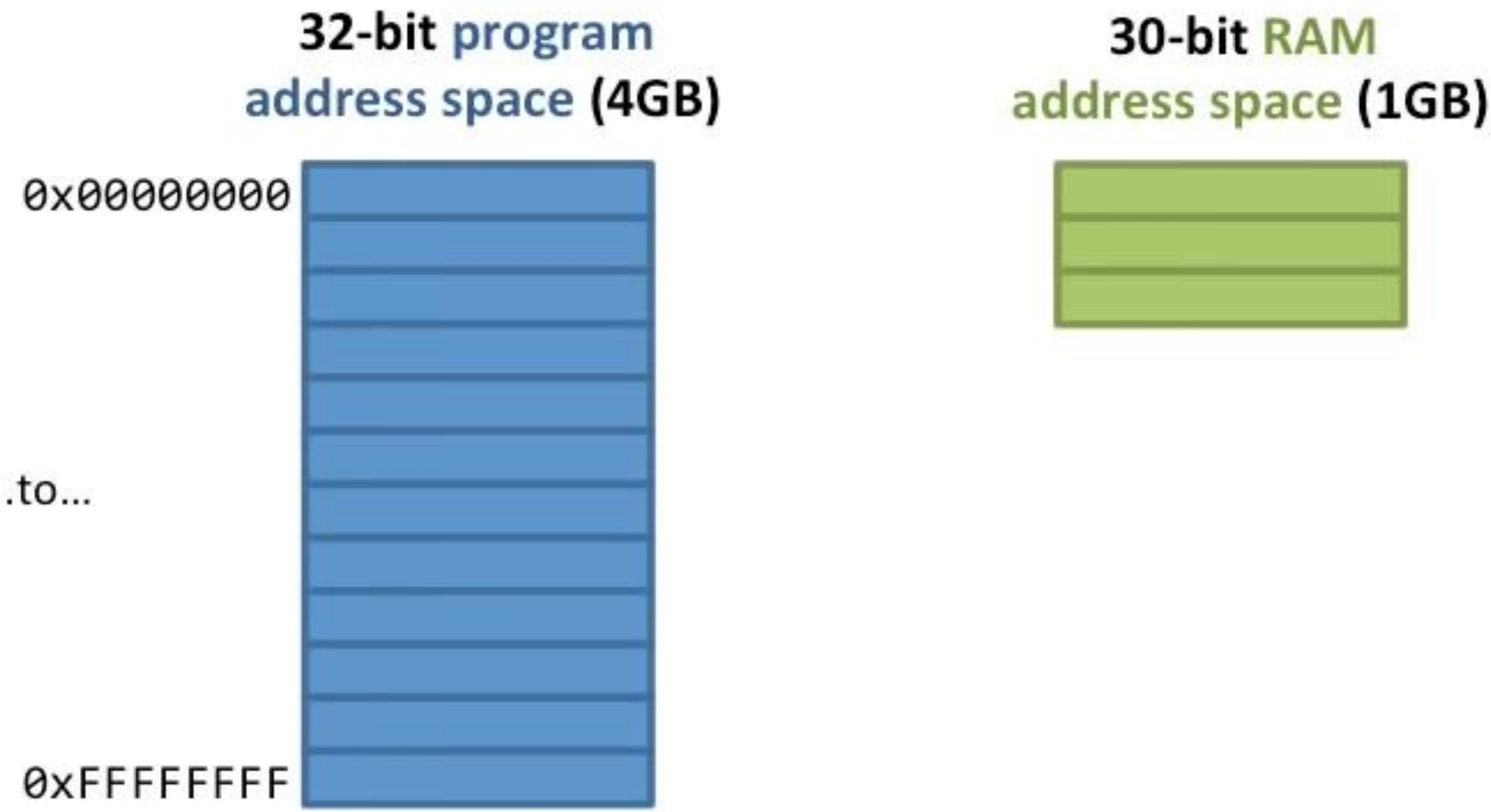
(2) Contiguous memory allocation

- Memory fragmentation

(3) Programs writing over each other

- STORE $1024, R1 in Program#1 and STORE $1024, R2 in Program#2
- Isolation Program

Understand The Problems: Not enough RAM



Understand The Problems: Contiguous memory allocation and Writing over

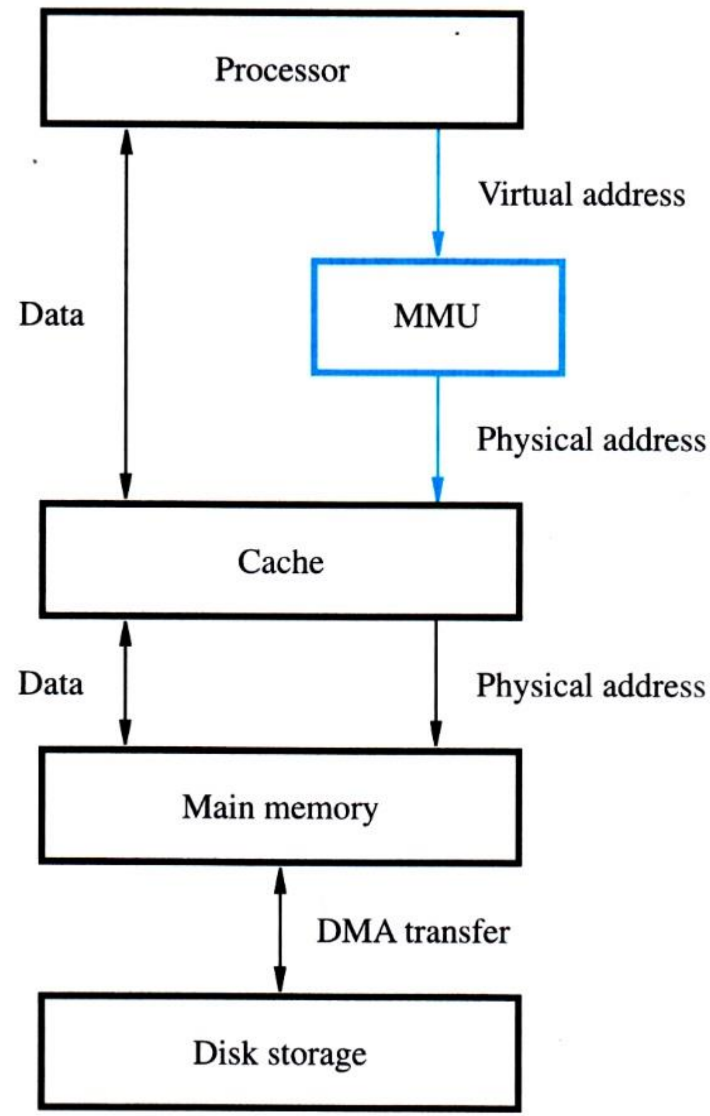
Virtual Memory

- **Give** each program, it's own virtual memory space.
- **Map** each program's virtual memory space to RAM's physical memory space.

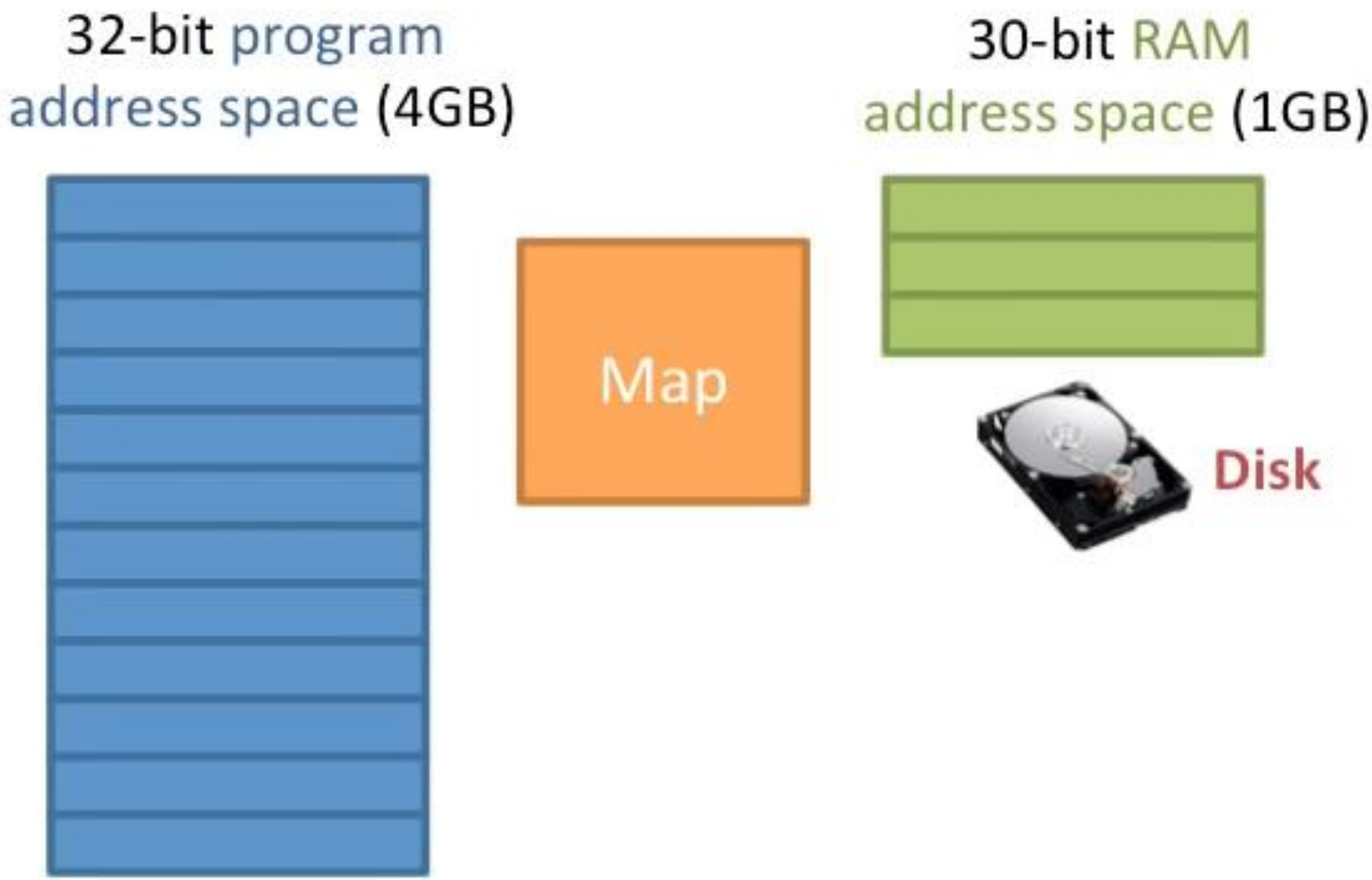
- Programmers will see the memory hierarchy as one big space.
- Programs will be able to access the virtual memory without any acknowledge of the physical hierarchy.
- The operating system manages the virtual memory.

Logical and Physical Address

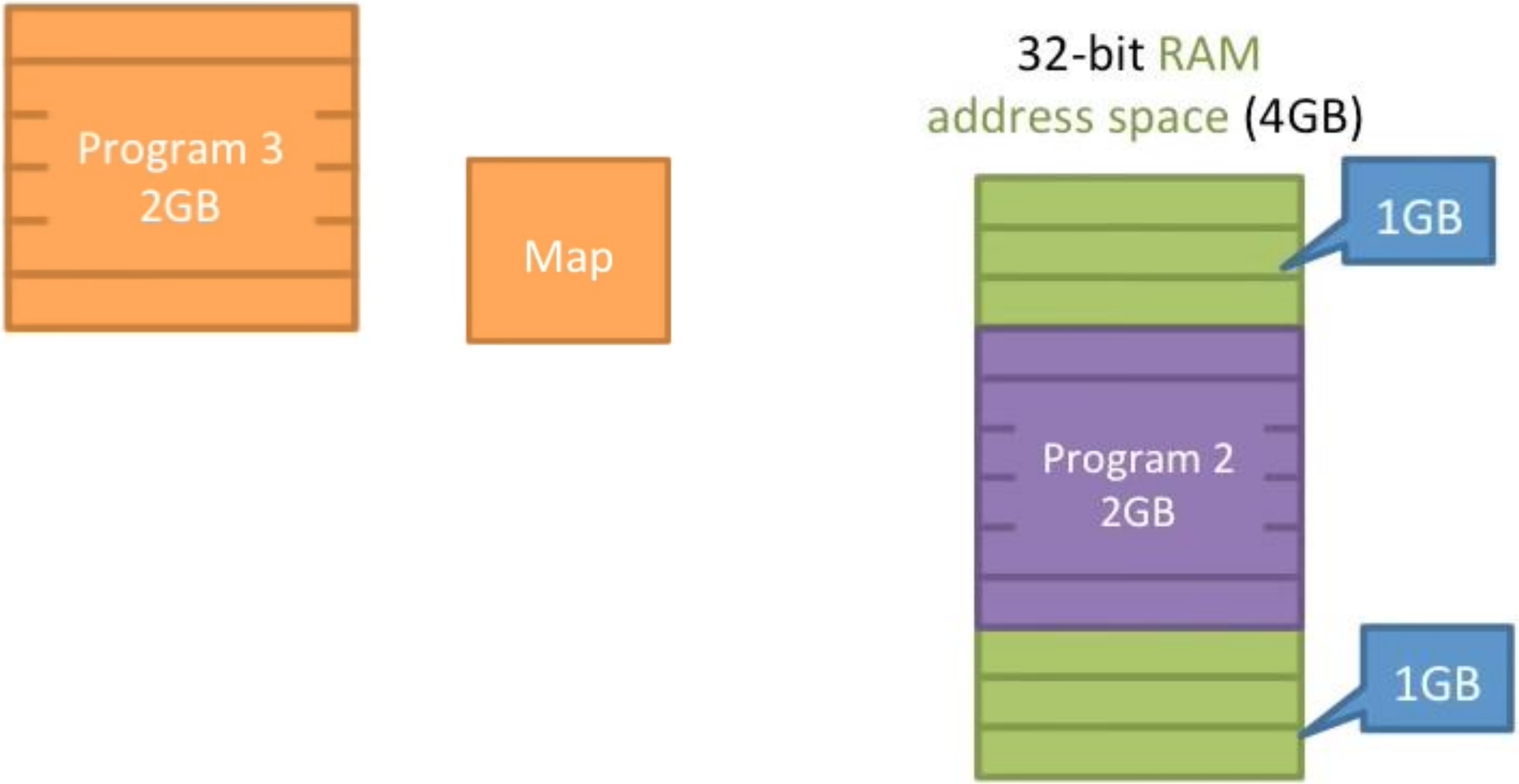
- Logical address, LA
  - or Virtual Address, VA
  - is the address that a program uses to refer a memory location.
- Physical address, PA
  - is the actual address.
- Memory Management Unit (MMU)
  - handles a virtual memory system. The MMU knows how to map from LA to PA or from PA to LA.



Understand GIVE&MAP



Understand How to Solve The Problems



Address Mapping

- A program refers to the logical address space (LA), when it is executed.
- The LA will be mapped to the physical address, (PA).
- The mapping is called *address binding* or *address translation*.
- The physical memory is a 1-D array of words, while the logical memory can be of any complex structure.
- Function $PA = f(LA)$ can be a complex function.

Understand Address Mapping

Logical Address Space

Processor
ld R3, $1024
ld R2, $00FF
add R1, R2, R3
ld R4, $0100
add R1, R1, R4

Address Mapping

MMU	
LA	PA
$0100	disk
$00FF	5
$1024	3

Physical Address Space

RAM

Disk

Block of word

Page Table

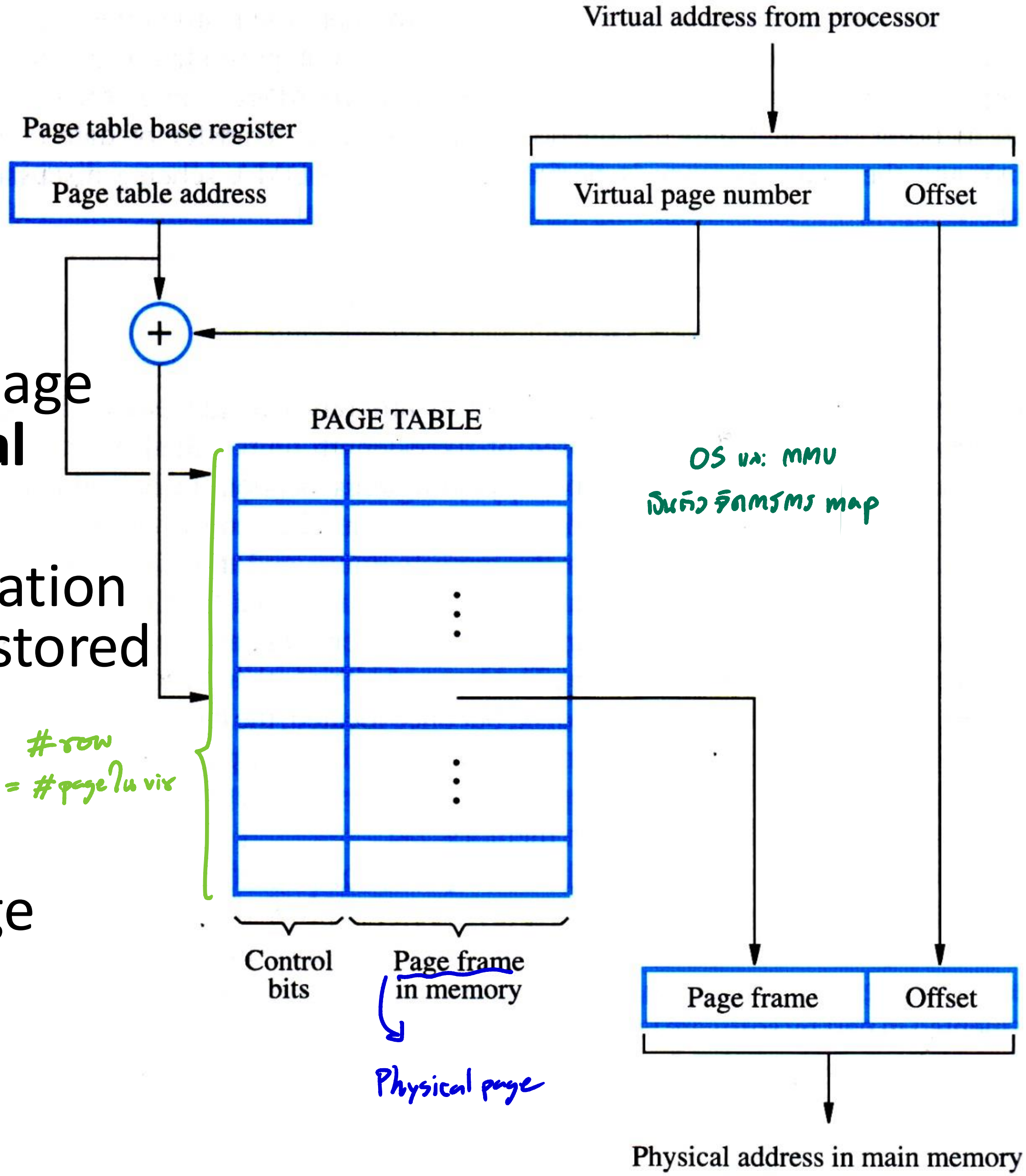
- How big of mapping table entry?

- Each block of fixed length is called a *page*
- A page size is about 2 to 16 KB.
  - If a page size is 2 MB.
- Data transfer between a memory device of different level is done by using a *page* as a basic unit.
- High order bit is a base address (page number) and low order bit is an offset indicating the position of a particular word in a page.

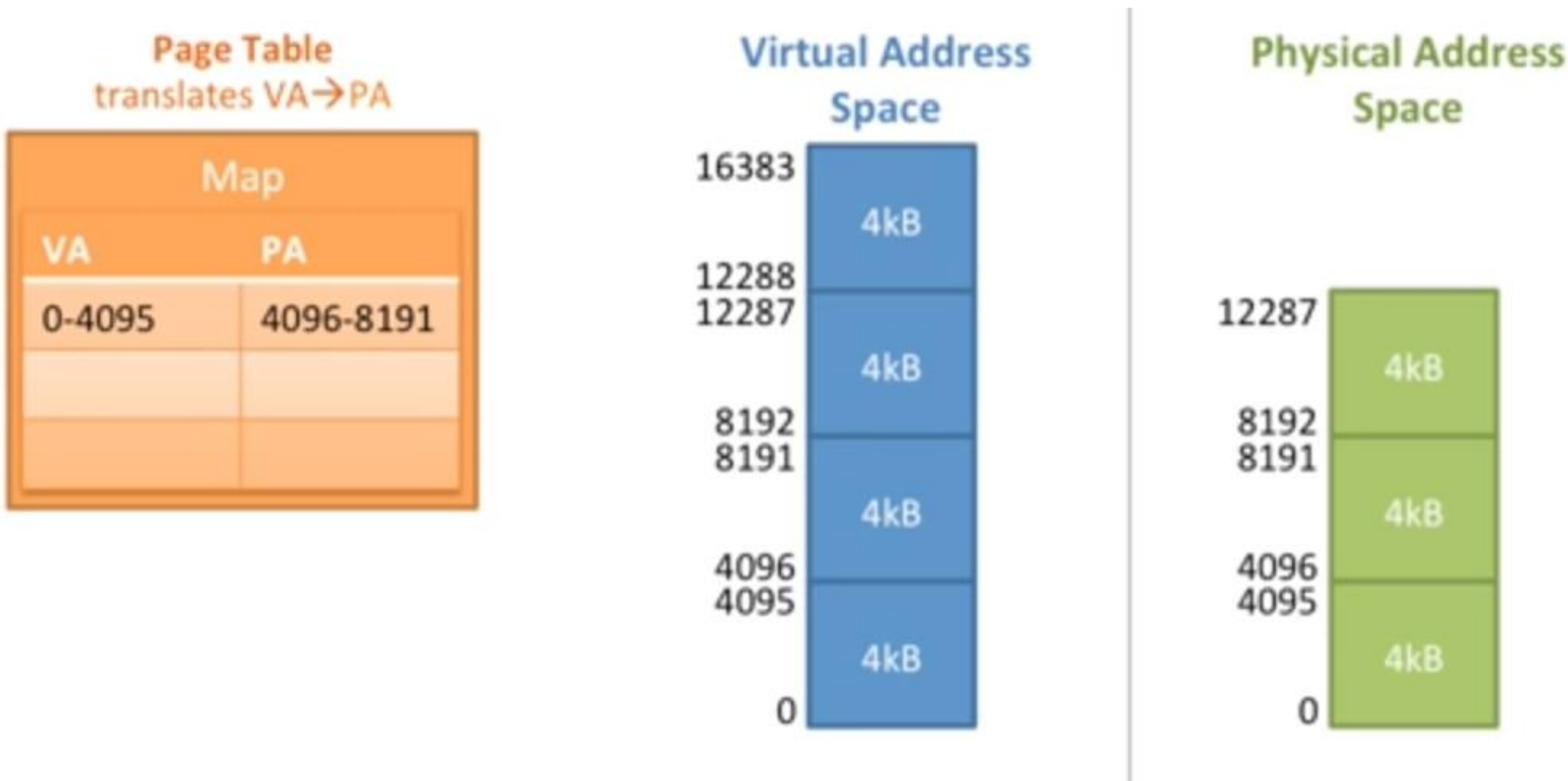
Address Translation

- A page table stores information about page position in **a physical memory**.
- The first address location of the page table is stored in a page table base register.
- The MMU uses information in a page table to access the memory.

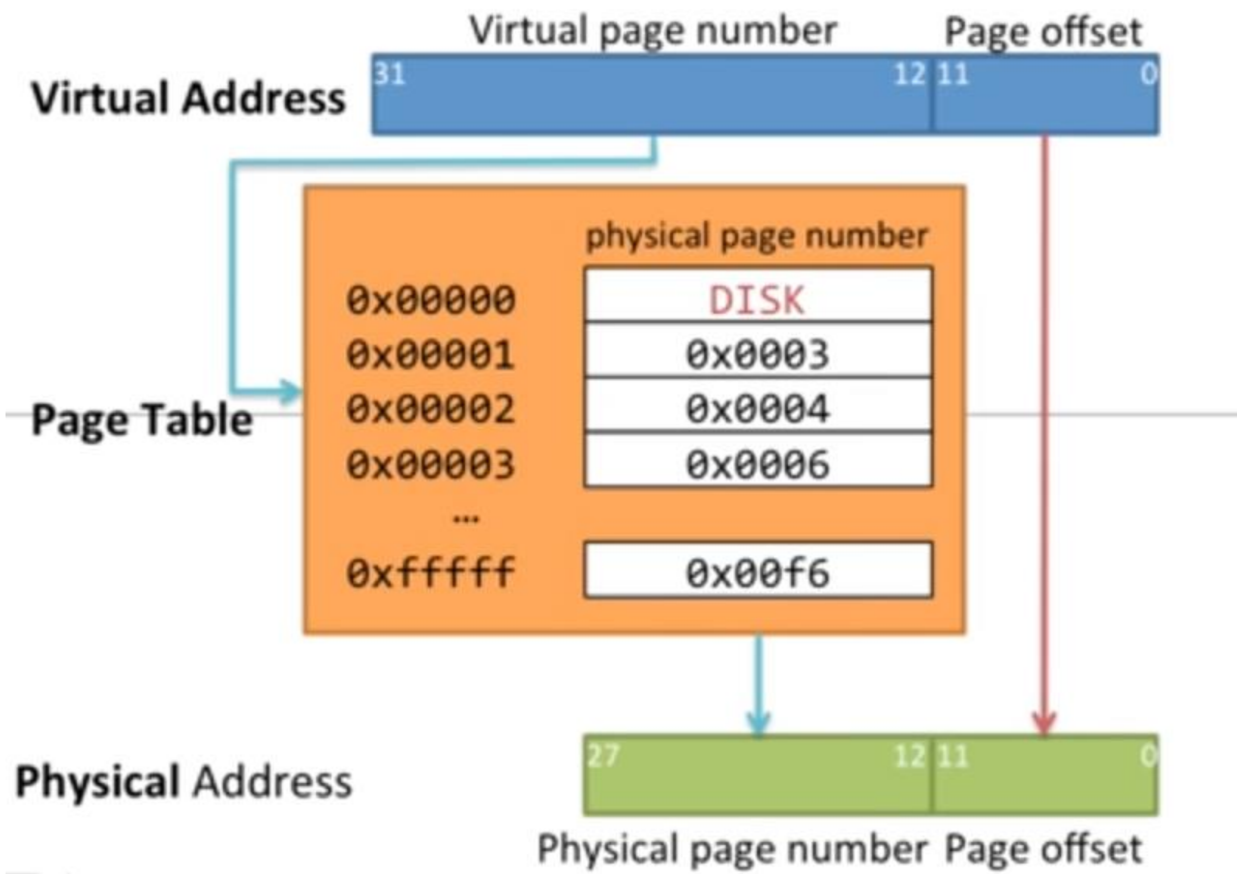
#row
= #page ใน vir



Draw This Together to Understand Page Mapping



Draw This Together to Understand Address Translation



Data ไม่อยู่ใน main mem
แต่อยู่ใน disk

Page Fault and Reduce Size of Page Table

ต้องดึงจาก disk
เข้า main mem ก่อน

See you on the next year

Page Selection and Replacement

Invert page table

Multi-level Page Table

Page Table เก็บใน RAM ต้อง access 2 รอบ

Cost of Address Translation

- Look up page table to translate the LA to PA, need to access page table in RAM รอบ 1
- Translate the address
- Access the data in RAM รอบ 2

2 main memory accesses

ทางแก้ : เก็บ page table ใน cache แทน

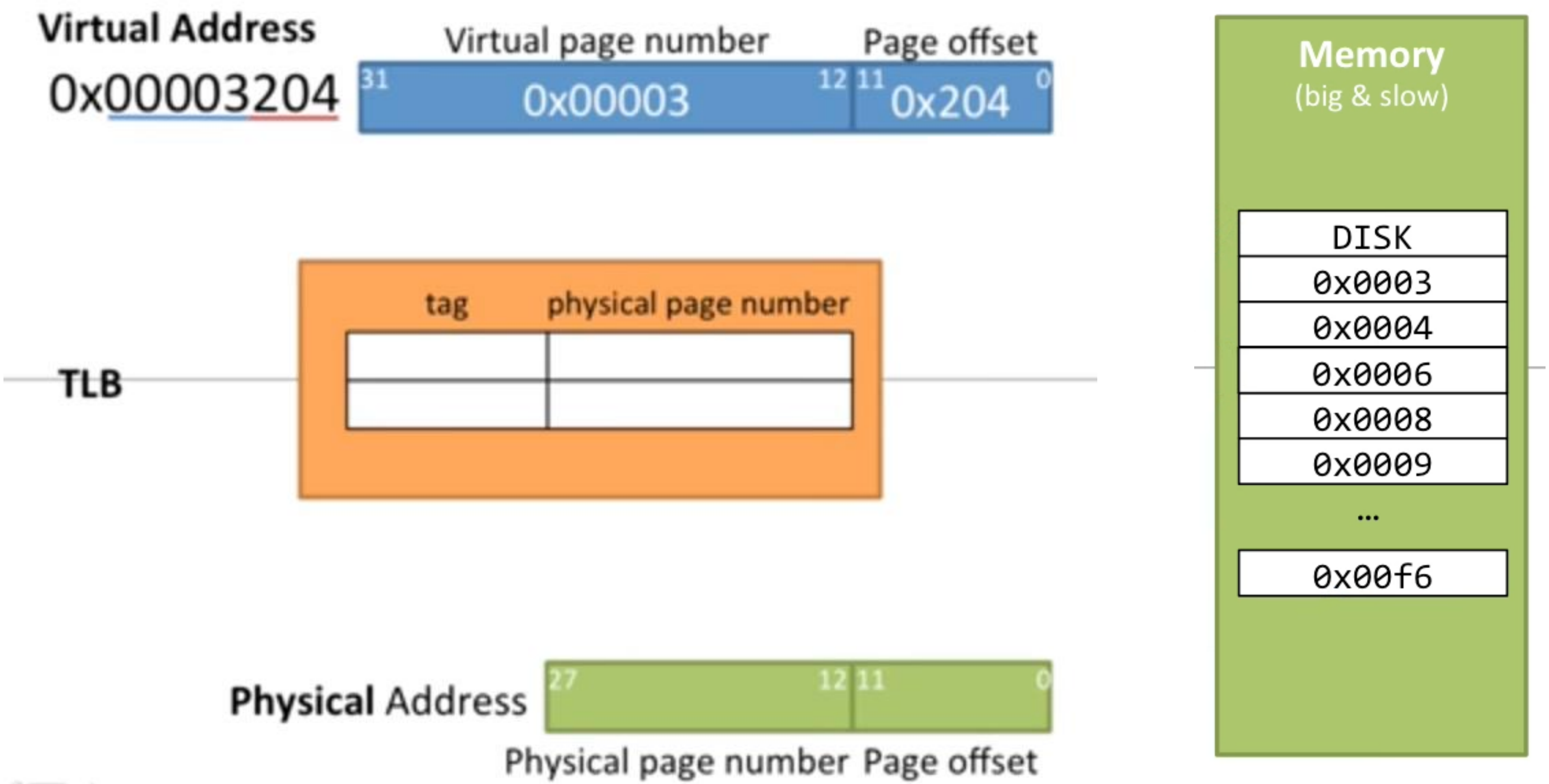
- Make a page table **cache**

Translation lookaside buffer (TLB)

- TLB is a special cache that has only one task, which is to translate memory address.
- The MMU tries to find a reference page frame in the TLB.
- If the page number is in the TLB, a physical address will be referred.
- If the page number is not in the TLB, cache miss will occur and memory transfer will take place.

หลักการต่อ cache เลย ยึดหลัก locality of reference
ถ้า page ถูกเข้าถึงแล้วก็เก็บใน cache ถ้ายังไม่เก็บใน cache ต้องไปดึงจาก RAM เข้า cache

Draw This Together to Understand Address Translation



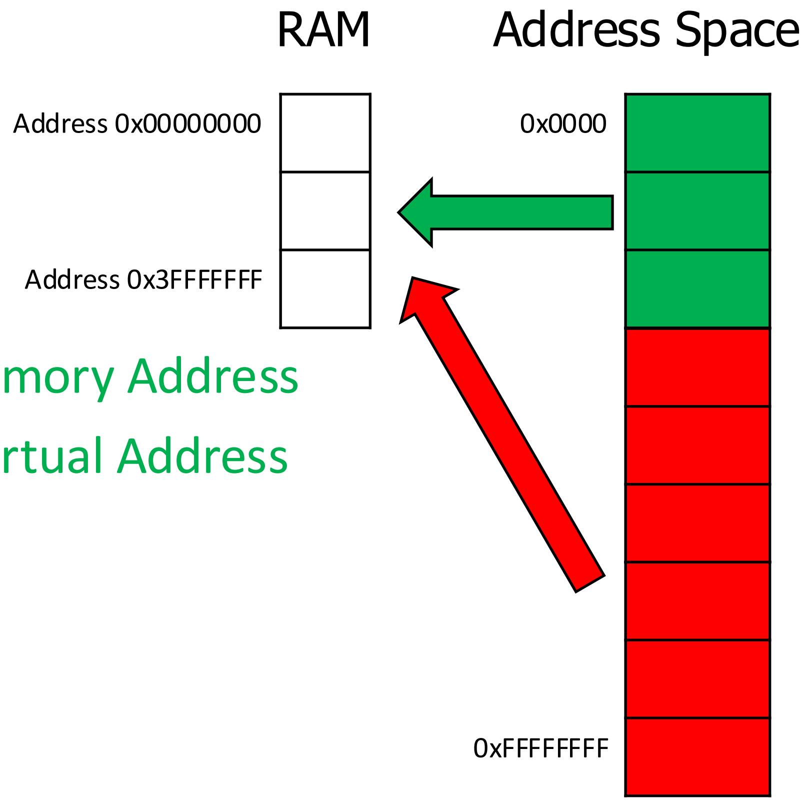
To be Faster with TLB

- Separate TLB for instruction (iTLB) and data (dTLB)
- Must TLB be small for easy to search
  - Just 64 entries from 1M entries of page table
- Make TLB seem bigger by increasing the page size
- Secondary TLB level
- Add H/W automatically fill page, called hardware page table walk

Supplementary Section

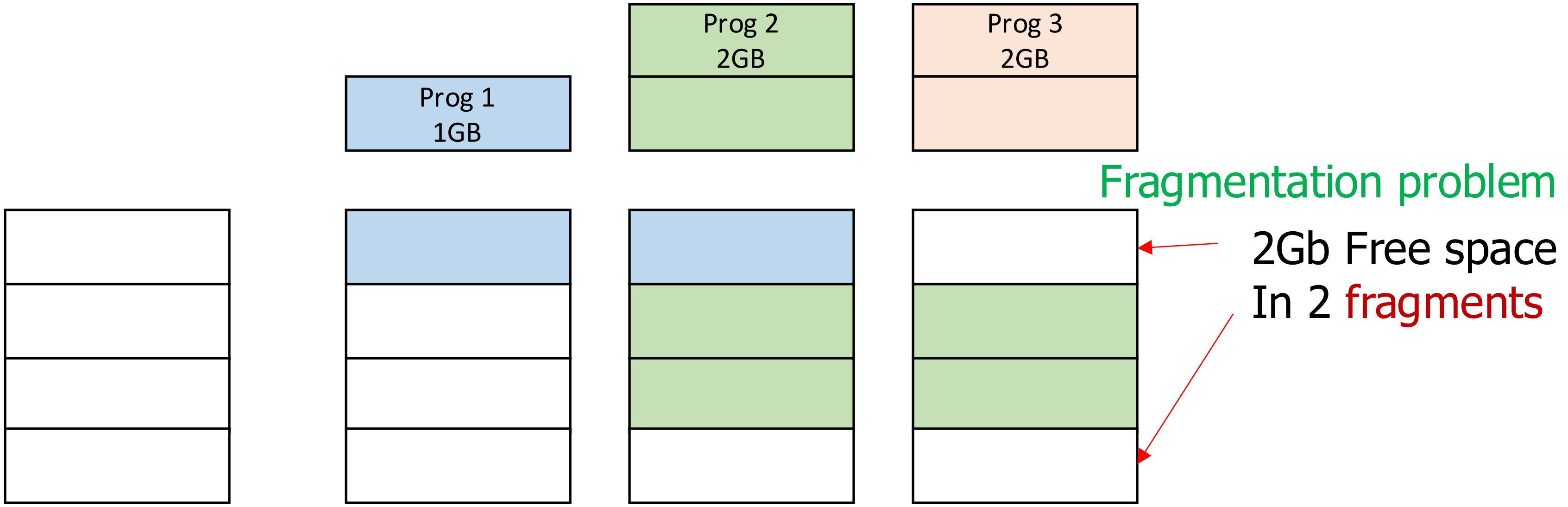
Not Enough RAM

- Suppose we use 32 Bit processor
- We only have 1GB of RAM
- The physical memory is smaller
- We have missing program address
- Crash if no virtual memory
- Have to put 4GB into 1 GB
- 1Gb of Physical Address or Main Memory Address
- 4GB of Program Address Space or Virtual Address



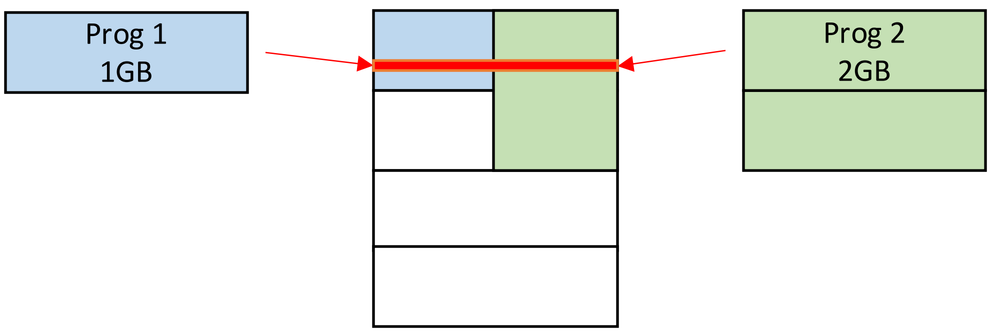
Holes in Address Space

- Have 4GB of Memory
- Have 3 Programs to run
- Program runs in contiguous address space
- Even though Prog1 is terminated, still cannot run Prog 3



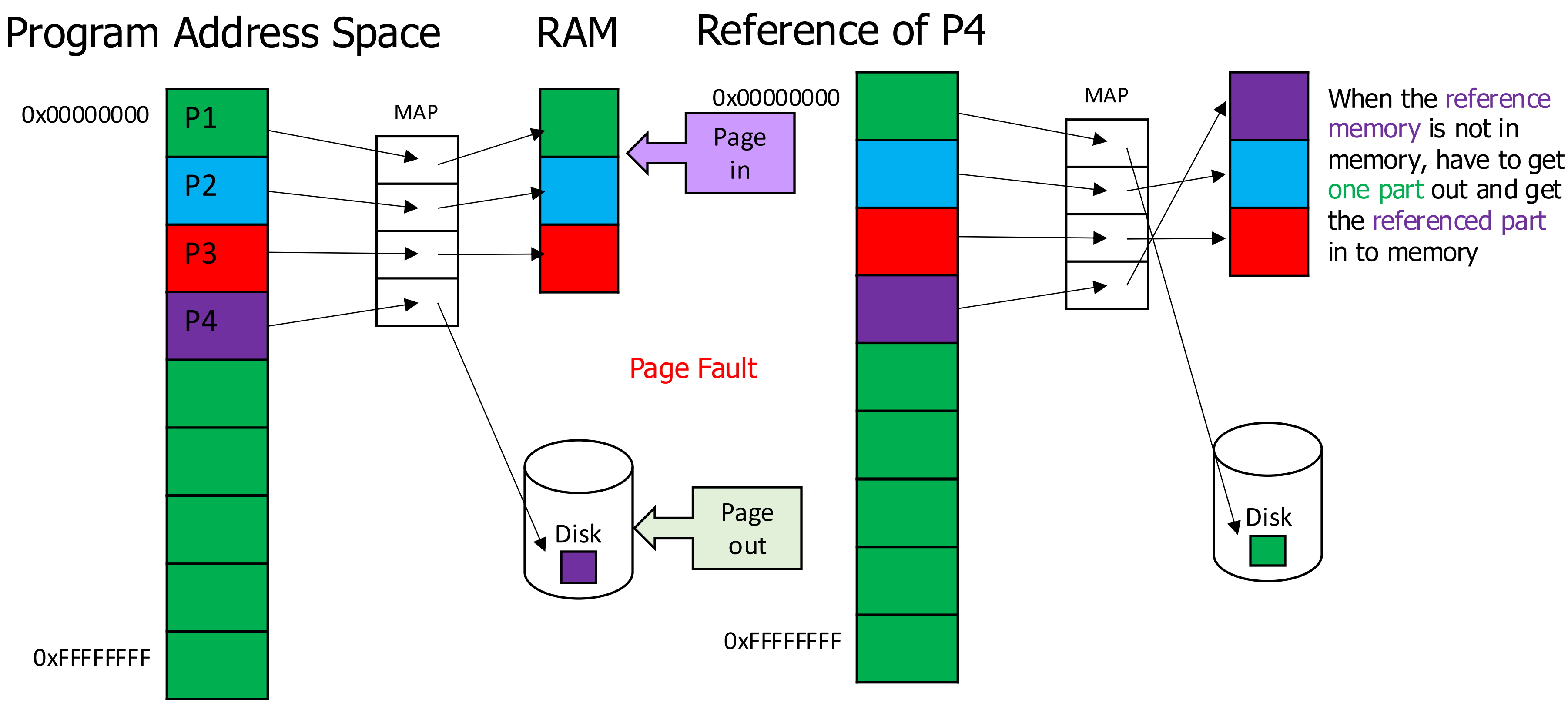
Overlapping Memory

- All programs can access 32-bit memory address
- Programs may access same memory address
- That could corrupt data or make program crashes



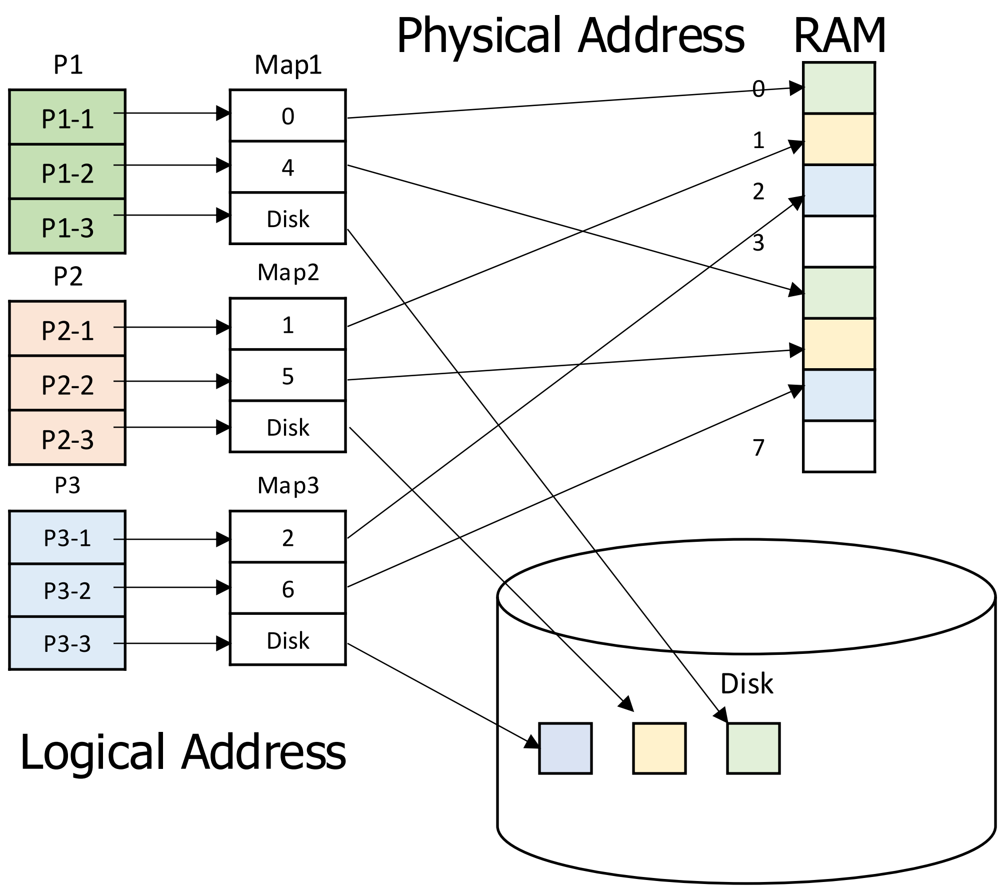
Virtual Memory: Mapping

- A map to add a level of indirection



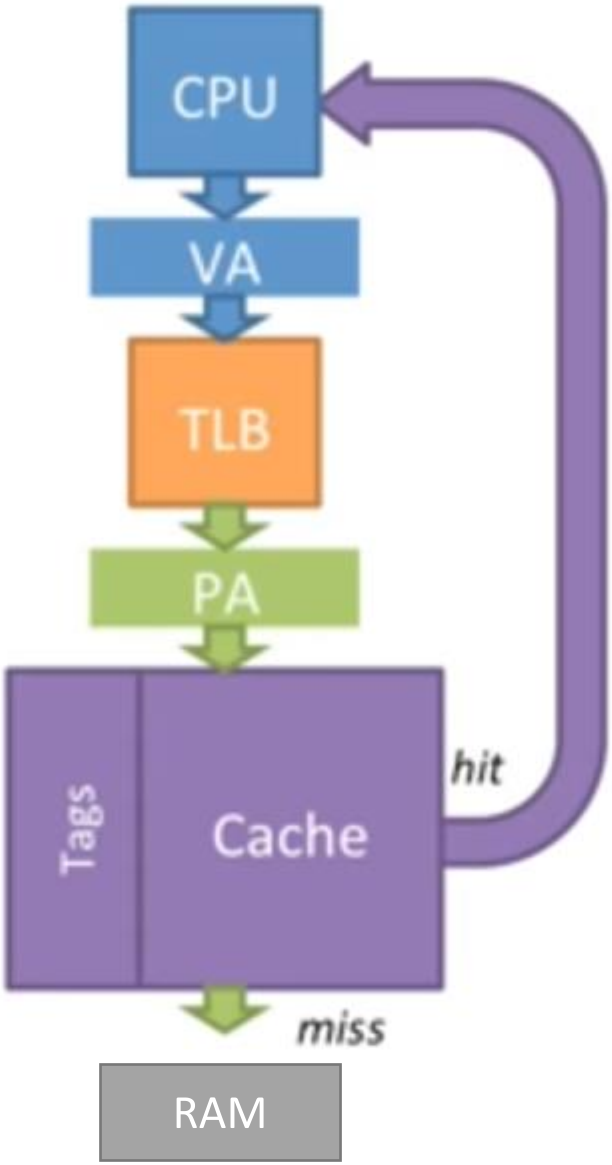
Virtual Memory: Multi-Program

- Program has its own map



No Overlapping
No Contiguous

Physical Cache



Virtual Cache